

A RACK ENGINEERED FOR A PART NOTHING ELSE FIT.

When an off-highway OEM had an oversized, awkward weldment with no standard rack to hold it, SPF's **in-house engineering** turned a rough idea into a buildable, ergonomic returnable rack.

1

DESIGN ROUND –
FIRST-PASS APPROVAL

1-3K LB

RATED LOAD
PER RACK

5-10

PARTS STAGED
PER RACK

THE CHALLENGE

An off-highway / heavy-equipment OEM needed to move large weldments — **[frames / axles / booms]** — between weld, machining, and assembly. The parts were heavy and awkward, no off-the-shelf rack fit, and manual handling raised damage and ergonomic concerns.

WHAT SPF DID

SPF's SolidWorks team took the part data and a rough concept and engineered a returnable rack around it — load paths, fork access, and operator ergonomics designed in. A prototype was reviewed, adjusted, and moved to production, all in-house.

THE RESULT

Approved on the **first design pass** and rated to 1,000–3,000 lb, the rack stages 5–10 parts safely, cut **[manual handling]**, and became the standard design for the program's follow-on parts.



CUSTOM-ENGINEERED RETURNABLE RACK · FOLDING STANCHIONS, MESH DECK

“[The rack did what three vendors said couldn't be done for that part.]”

[MANUFACTURING ENGINEER, HEAVY-EQUIPMENT OEM]

WHY SPF

BUILD-TO-PRINT

IN-HOUSE DESIGN

ROBOTIC WELDING

ONE ROOF

INDUSTRIES

HEAVY EQUIPMENT

INDUSTRIAL MACHINERY

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